MBFT 814 : Microbial Biotechnology 3(2-0-2)

M.Sc. Microbiology, Semester III · Detailed mid-term notes

Anupama Pandey · P.I.D. 25MSCMB010 · SHUATS, Prayagraj

#	Topic (mid-term syllabus)	Main points covered
1	Introduction to microbial biotechnology	Definitions, why microbes, industrial strain, fermentation process components, media, products, scope
2	History of microbial biotechnology	Eras, detailed timeline, key discoveries explained, Indian milestones
3	Types of bioreactors	Parts with labelled diagram, ideal bioreactor, CSTR, bubble column, airlift, fluidized bed, packed bed, membrane, photobioreactor
4	Types of fermentation methods	Batch (growth curve, kinetics), fed-batch, continuous (chemostat, turbidostat), SmF, surface, SSF, semi-solid
5	Enzyme production	Types, regulation, steps, amylase and protease examples, downstream flow chart, units, applications
6	Immobilization of enzymes	History, supports, 5 methods with procedure and diagram, properties, reactors, applications
7	Screening and strain improvement	Primary and secondary screening, mutagens and their action, mutant selection, protoplast fusion, rDNA
8	Antibiotic production	Classification; penicillin (biosynthesis, process, flow chart); streptomycin; tetracycline; griseofulvin
9	Vitamin production	Riboflavin (B_2), cyanocobalamin (B_{12}), vitamin C, β-carotene
10	Quick revision and likely questions	One-line facts and question bank

1. INTRODUCTION TO MICROBIAL BIOTECHNOLOGY

1.1 Definitions

- **Biotechnology**: the word was first used by the Hungarian engineer **Karl Ereky in 1919** for the production of products from raw materials with the help of living organisms.
- **OECD / Convention on Biological Diversity (1992)**: "Any technological application that uses biological systems, living organisms or derivatives thereof, to make or modify products or processes for specific use."
- **European Federation of Biotechnology**: "the integrated use of **biochemistry, microbiology and engineering sciences** in order to achieve technological (industrial) application of the capabilities of microorganisms, cultured tissue cells and parts thereof".
- **Microbial biotechnology**: the branch that uses **microorganisms** (bacteria, actinomycetes, fungi, yeasts, microalgae), their enzymes or their genes to make useful products (food, drugs, chemicals, fuels) or to give services (waste treatment, bioremediation) on a commercial scale.
- **Fermentation (industrial meaning)**: any process in which microorganisms are grown on a large scale to form a product, whether aerobic or anaerobic. (In the biochemical meaning, fermentation is anaerobic energy-yielding metabolism where organic compounds act as electron donor and acceptor.)

1.2 Why microorganisms are used

Property	Why it matters in industry
Small size, large surface : volume ratio	Very fast uptake of nutrients and release of products, so high metabolic rate
Rapid growth (generation time 20 min to a few hours)	Large biomass in hours or days instead of months (plants, animals)
Metabolic diversity	Can use sugars, starch, cellulose, hydrocarbons, methanol, CO_2; make acids, alcohols, enzymes, antibiotics, vitamins, polymers
Grow on cheap raw materials	Molasses, corn steep liquor, whey, bran, agro-industrial wastes lower the cost
Easy genetic manipulation	Mutation and recombinant DNA can raise yield many times
Controlled production	Independent of climate, season and land; uniform quality; easy scale-up in fermenters

1.3 Characteristics of an ideal industrial microorganism

- Pure culture, genetically **stable**, and easy to preserve for a long time.
- Gives a **high yield** of the required product in a **short time**.
- Grows on **cheap**, easily available medium; forms few unwanted by-products.
- Non-pathogenic, non-toxic, preferably **GRAS** (Generally Regarded As Safe).
- Produces spores or cells that are easy to use as **inoculum**.
- Product is easy to separate (preferably **extracellular**); resistant to contamination and phage.

1.4 Components (stages) of a fermentation process

Every industrial fermentation has six basic parts (Stanbury, Whitaker and Hall):

1. **Formulation of media** for the inoculum and production stages.
2. **Sterilization** of the medium, fermenter and air (medium: batch at 121°C, or continuous high-temperature short-time; air: filtration).
3. **Inoculum development**: stock culture → slant → shake flask → seed fermenter → production fermenter (usually 3–10% v/v inoculum).
4. **Production** in the fermenter under optimum pH, temperature, aeration and agitation.
5. **Downstream processing**: separation, extraction, purification and formulation of the product.
6. **Effluent treatment**: disposal of spent broth and biomass.

Steps 1–4 are called **upstream processing**; step 5 is **downstream processing**.

1.5 Fermentation media

Component	Examples	Role
Carbon source	Glucose, sucrose, **molasses**, starch, lactose, whey, vegetable oils, methanol	Energy and carbon skeleton
Nitrogen source	**Corn steep liquor (CSL)**, soybean meal, yeast extract, peptone, ammonium salts, urea	Proteins, nucleic acids
Minerals	P, K, Mg, S, trace Fe, Zn, Mn, Co, Cu	Enzyme cofactors, cell structure
Growth factors	Vitamins, amino acids (often supplied by CSL, yeast extract)	Needed by fastidious strains
Precursors	Phenylacetic acid (penicillin G), glycine (riboflavin), cobalt + DMB (vitamin B_{12})	Built directly into the product, raise yield
Inducers	Starch (amylase), oils (lipase), cellulose (cellulase)	Switch on enzyme genes
Buffers / antifoams	$CaCO_3$, phosphates / silicone oils, vegetable oils	Control pH / control foam

1.6 Types of microbial products

- **Microbial cells (biomass)**: baker's yeast, single cell protein (SCP), biofertilizers, probiotics, vaccines.
- **Microbial enzymes**: amylase, protease, lipase, cellulase, glucose isomerase.
- **Primary metabolites**: ethanol, citric acid, lactic acid, amino acids, nucleotides, vitamins.
- **Secondary metabolites**: antibiotics, pigments, toxins, alkaloids, gibberellins.
- **Recombinant products**: insulin, interferons, growth hormone, hepatitis B vaccine.
- **Biotransformation products**: steroid conversions (progesterone to 11α-hydroxyprogesterone by *Rhizopus*), vitamin C step by *Gluconobacter*.

Primary metabolites	Secondary metabolites
Made during active growth: **trophophase** (log phase)	Made after growth slows: **idiophase** (late log / stationary)
Essential for growth and reproduction	Not essential for growth; help survival (competition, defence)
Common to most organisms; produced in the same way	Restricted to a few species or strains
Production is growth-linked	Production is non-growth-linked; often repressed by glucose, phosphate
e.g. ethanol, citric acid, amino acids, nucleotides, vitamins	e.g. penicillin, streptomycin, tetracycline, pigments, mycotoxins

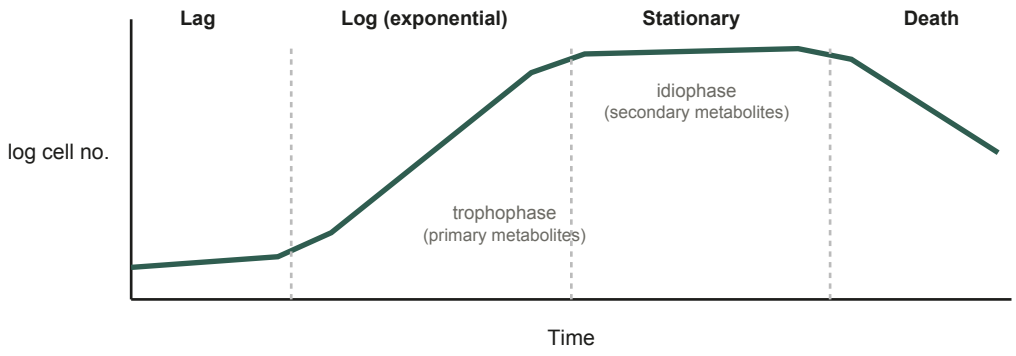


Fig. 1. Batch growth curve showing trophophase and idiophase

1.7 Scope and applications

Area	Examples
Food and beverages	Beer, wine, bread, cheese, curd, vinegar, soy sauce, SCP, amino acids (glutamic acid as MSG, lysine), flavours

Area	Examples
Health and pharmaceuticals	Antibiotics, vitamins, vaccines, insulin, interferons, monoclonal antibodies, steroids, therapeutic enzymes
Industrial chemicals	Ethanol, acetone, butanol, glycerol, citric, lactic, gluconic and itaconic acids, biopolymers (PHB, xanthan)
Enzymes	Detergent, starch, textile, leather, paper, dairy, juice industries
Agriculture	Biofertilizers (*Rhizobium*, *Azotobacter*, PSB), biopesticides (*Bacillus thuringiensis*, *Trichoderma*), silage, plant growth promoters
Environment	Sewage and effluent treatment, bioremediation of oil and pesticides, biogas, composting, bioleaching of copper and uranium
Energy	Bioethanol, biogas (methane), biodiesel, biohydrogen, microbial fuel cells

Exam point: For a long answer: definitions (Ereky 1919 + OECD) → why microbes → ideal industrial strain → six components of a fermentation process → primary vs secondary metabolites with growth curve → scope table.

2. HISTORY OF MICROBIAL BIOTECHNOLOGY

2.1 Eras of biotechnology

Era	Period	Features
Ancient biotechnology	Before 2000 BC (8000–4000 BC)	Domestication of plants and animals; fermented foods made by chance without knowing microbes
Classical biotechnology	2000 BC to 1900s AD	Deliberate fermentation for food and medicine; later scientific understanding of microbes (Leeuwenhoek, Pasteur, Koch)
Industrial / antibiotic era	1900–1953	Acetone–butanol, citric acid, penicillin, streptomycin; deep-tank fermentation; genetics
Molecular era	1953–1976	DNA structure, genetic code, restriction enzymes
Modern biotechnology	1977 onwards	Genetic engineering, recombinant proteins, PCR, genomics, metabolic engineering

2.2 Ancient and classical period

- **About 6000 BC**: Sumerians and Babylonians brewed **beer** with yeast.
- **About 4000 BC**: Egyptians made **leavened bread** using yeast; wine, vinegar and cheese were also made. Cheese began among nomadic tribes about 4000 years ago using **rennet** from calf stomach.
- Food was preserved by drying, salting, smoking and by fermentation (the acid formed stops spoilage).
- **1673–1676**: **Antonie van Leeuwenhoek**, with his single-lens microscope, first described bacteria and protozoa ("animalcules").
- **1701**: Giacomo Pylarini practised inoculation against smallpox; **1796**: **Edward Jenner** used cowpox to vaccinate against smallpox.
- **1809**: **Nicolas Appert** developed canning (heat sterilization of food in sealed containers).

2.3 The golden age of microbiology (1850–1900)

- **Louis Pasteur** (1822–1895), "father of industrial microbiology": showed in **1857** that lactic acid fermentation and later alcoholic fermentation are caused by **living microorganisms**; different microbes give different products; solved "diseases" of wine and beer; introduced **pasteurization**; described the **Pasteur effect** (oxygen inhibits fermentation); disproved spontaneous generation (swan-neck flask, 1861).
- **Robert Koch**: pure culture techniques on solid media and Koch's postulates (1876–1884), which made pure-culture industrial fermentation possible.
- **1877**: Wilhelm Kühne coined the word **"enzyme"** (Greek, "in yeast").
- **1897**: **Eduard Buchner** showed that a **cell-free yeast extract** ("zymase") could ferment sugar to alcohol. This proved fermentation is due to **enzymes**, not the "vital force" of living cells; Nobel Prize 1907.

2.4 Industrial fermentation era (1900–1950)

- **1914–1918**: **Chaim Weizmann** developed the **acetone–butanol–ethanol (ABE) fermentation** using *Clostridium acetobutylicum* on starch (maize). Acetone was needed to make **cordite** (explosive) during World War I. This was the first large-scale industrial fermentation needing strict aseptic conditions.
- **1917–1923**: J. N. Currie showed *Aspergillus niger* makes **citric acid** in surface culture; **Pfizer** started commercial production in 1923.
- **1919**: **Karl Ereky** coined "biotechnology".
- **1928**: **Alexander Fleming** noticed that *Penicillium notatum* killed *Staphylococcus* colonies and named the substance penicillin.
- **1938–1941**: **Howard Florey and Ernst Chain** (Oxford) purified penicillin and showed its clinical value (Nobel 1945 with Fleming).

- **1941–1945**: At the **Northern Regional Research Laboratory, Peoria (USA)**, penicillin production moved from surface flasks to **submerged deep-tank fermentation**: **corn steep liquor** raised yield, and the high-yielding strain *P. chrysogenum* **NRRL 1951** was found on a mouldy cantaloupe. This created modern fermentation technology (aeration, agitation, sterile air).
- **1944**: **Selman Waksman**, Albert Schatz and Elizabeth Bugie discovered **streptomycin** from *Streptomyces griseus*, the first drug effective against tuberculosis (Nobel 1952). Waksman coined the word "antibiotic".
- **1947–1950**: chloramphenicol (*S. venezuelae*), chlortetracycline (1948, Duggar, *S. aureofaciens*), and other antibiotics: "golden age of antibiotics".

2.5 1950–1976

- **1950s**: microbial **biotransformation of steroids** (e.g. 11α-hydroxylation of progesterone by *Rhizopus*, 1952) made cortisone cheap.
- **1953**: **Watson and Crick** proposed the double-helix structure of DNA.
- **1957**: **Kinoshita** (Japan) produced **glutamic acid** with *Corynebacterium glutamicum*: start of amino acid fermentation.
- **1960s–1970s**: **single cell protein** (SCP) from hydrocarbons and methanol (ICI "Pruteen" with *Methylophilus methylotrophus* in a large airlift fermenter); industrial enzymes such as glucose isomerase.
- **1969**: first industrial use of an **immobilized enzyme** (aminoacylase, Tanabe Seiyaku, Japan, I. Chibata).
- **1970**: discovery of **restriction enzymes** (Hamilton Smith); **1972**: Paul Berg made the first recombinant DNA molecule; **1973**: **Cohen and Boyer** cloned DNA in *E. coli* using a plasmid.

2.6 Modern biotechnology (1977 onwards)

Year	Event
1975	Köhler and Milstein: hybridoma technique for monoclonal antibodies
1976	Genentech founded (first genetic engineering company)
1977	Genentech produced **somatostatin** in *E. coli*: first human protein from a cloned gene
1978	Human **insulin** gene expressed in *E. coli*; rat insulin made by rDNA at Harvard
1982	**Humulin** (recombinant human insulin) approved: first rDNA drug
1983	Kary Mullis invented **PCR**
1986	First recombinant vaccine: **hepatitis B** surface antigen made in yeast; India set up the **Department of Biotechnology (DBT)**
1990–2003	Human Genome Project
1995	First complete bacterial genome: *Haemophilus influenzae*
1998	Remicade (monoclonal antibody for Crohn's disease) approved
2012 onward	CRISPR-Cas9 genome editing (Doudna and Charpentier); synthetic biology and metabolic engineering

Indian milestones: Hindustan Antibiotics Ltd., Pimpri (1954) for penicillin; IDPL (1961); DBT (1986); India is now a major producer of vaccines and generic antibiotics.

Exam point: Write history as eras with names and years. Must-know: Leeuwenhoek 1676, Pasteur 1857, Buchner 1897, Weizmann WWI, Ereky 1919, Fleming 1928, Florey–Chain and Peoria 1940s, Waksman 1944, Watson–Crick 1953, Cohen–Boyer 1973, somatostatin 1977, insulin 1982.

3. TYPES OF BIOREACTORS

3.1 Definition and functions

A **bioreactor** or **fermenter** is a vessel in which microorganisms, animal or plant cells, or enzymes are grown or used under **controlled, aseptic** conditions to carry out a biochemical process. Its function is to give the best environment (temperature, pH, dissolved oxygen, nutrients, mixing) so that the organism forms the maximum amount of product.

3.2 Requirements of an ideal bioreactor

- Able to run **aseptically** for long periods (sterilizable by steam in place).
- Good **aeration and agitation** to meet oxygen demand, without damaging cells (low shear).
- Low power consumption; **temperature control** (jacket or coils); **pH control**.
- Sampling, feeding and harvesting facilities; minimum evaporation loss.
- Foam control; easy cleaning and maintenance; made of **non-toxic, corrosion-resistant** material (stainless steel grade 316, glass for small units).
- Can be scaled up.

3.3 Parts of a stirred tank fermenter

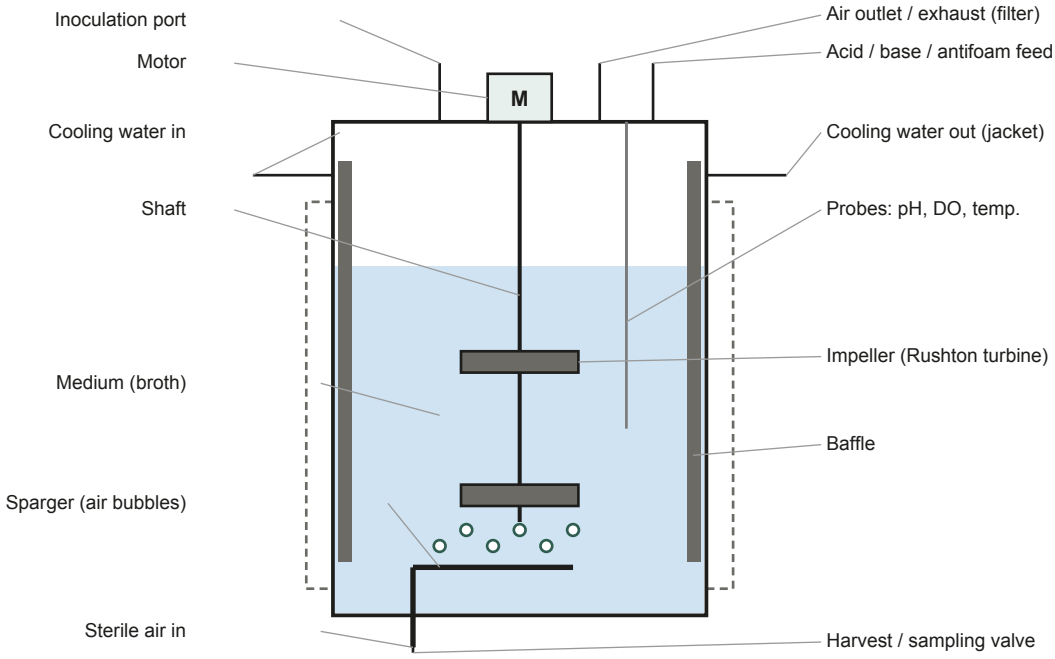


Fig. 2. Labelled diagram of a continuous stirred tank (stirred tank) fermenter

Part	Description and function
Vessel	Stainless steel cylinder with dished top and bottom; height : diameter about 2–3 : 1; working volume about 75–80% (headspace for foam)
Agitator (impeller)	Mounted on a shaft driven by a motor. **Rushton disc turbine** (flat blades on a disc) gives radial flow and breaks bubbles; marine propeller and pitched-blade turbines give axial flow. Mixes cells, nutrients and air and improves O_2 transfer
Sparger	Introduces sterile air at the bottom, below the impeller. Types: **porous** (sintered), **orifice** (perforated pipe or ring) and **nozzle** (open pipe)
Baffles	Usually **4 vertical metal strips**, width about 1/10 of vessel diameter, fixed to the wall; prevent **vortex** formation and improve mixing and aeration
Jacket / coils	Cooling or heating water circulates to keep temperature constant (removes metabolic and agitation heat)

Part	Description and function
Aeration system	Air compressor → sterile air filter → sparger; exhaust air through filter/condenser. Oxygen transfer is measured as $k_L a$
Foam control	Foam sensor with automatic addition of **antifoam** (silicone oils, vegetable oils) or mechanical foam breaker
Control probes	Temperature (thermocouple), **pH electrode** with acid/base pumps, **dissolved oxygen (DO) electrode**, pressure gauge, level sensor
Ports and valves	Inoculation, sampling, feeding (nutrients, precursor), and harvest valve; steam lines for sterilization in place

3.4 Types of bioreactors

(a) Continuous stirred tank reactor (CSTR)

- **Construction:** cylindrical vessel with motor-driven central shaft carrying one or more impellers, baffles, sparger, jacket (Fig. 2). Number of impellers depends on height.
- **Working:** air from the sparger is broken into small bubbles by the impeller and spread through the broth, giving a **uniform** environment.
- **Advantages:** efficient gas transfer, good mixing, flexible, easy control; the most widely used type.
- **Disadvantages:** high power use, high **shear** can damage filamentous fungi and animal cells, sealing of the shaft is a contamination risk.
- **Uses:** antibiotics, enzymes, amino acids, most aerobic fermentations.

(b) Bubble column bioreactor

- **Construction:** tall cylindrical column (height : diameter over 3 : 1) with a sparger at the base; **no mechanical agitator**; perforated plates may be placed inside to redistribute bubbles.
- **Working:** rising air bubbles both aerate and mix the medium.
- **Advantages:** simple, cheap, low shear, no moving parts, low contamination risk, low energy.
- **Disadvantages:** poor mixing at low gas flow, foaming, not good for viscous broth.
- **Uses:** baker's yeast, beer, vinegar, SCP, wastewater.

(c) Airlift bioreactor

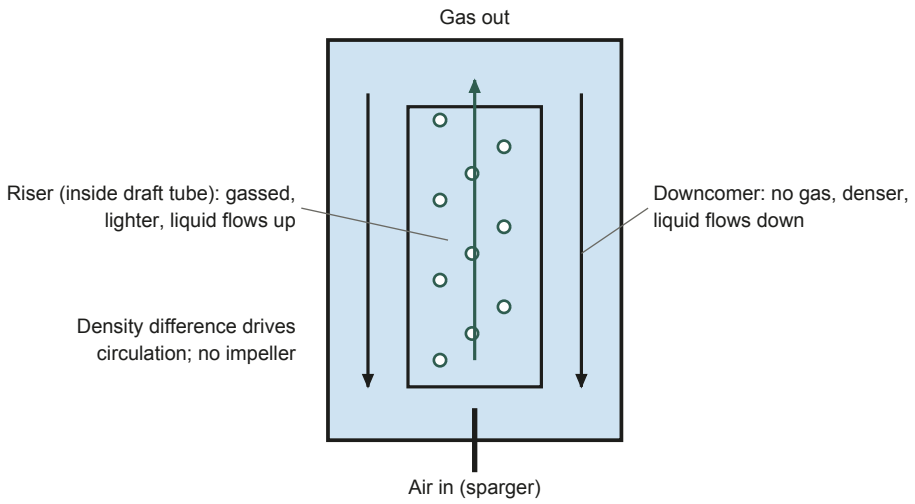


Fig. 3. Internal-loop airlift bioreactor (riser and downcomer)

- **Construction:** the vessel is divided into two zones by a **draft tube** or baffle: the **riser** (air is sparged into it) and the **downcomer** (no air). **Internal-loop** type has the draft tube inside; **external-loop** type has a separate outside tube connecting top and bottom.
- **Working:** the gassed liquid in the riser is less dense and rises; at the top gas escapes; the denser degassed liquid falls in the downcomer. This **density difference** causes circulation without any impeller.

- **Advantages:** low shear (good for plant and animal cells, fungi), low energy, good mixing and O_2 transfer, simple, easy to scale up, low contamination risk.
- **Disadvantages:** needs high air flow, less suitable for very viscous broths, foaming.
- **Uses:** SCP (ICI Pruteen plant, about 1500 m^3), plant and animal cell culture, monoclonal antibodies, wastewater treatment.

(d) Fluidized bed bioreactor

- **Construction:** similar to a bubble column but with a **narrow reaction column** and an **expanded top** section where fluid velocity falls, so particles fall back and are retained.
- **Working:** medium is pumped **upward** through a bed of small particles carrying immobilized cells or enzymes, fast enough to keep them **suspended (fluidized)**; liquid is often recycled.
- **Advantages:** good mixing and mass transfer, no clogging (unlike packed bed), suits immobilized cells, continuous operation.
- **Disadvantages:** hard to design and scale up, particles can be damaged.
- **Uses:** wastewater treatment (anaerobic), beer and vinegar with immobilized cells, immobilized enzyme reactions.

(e) Packed bed bioreactor

- **Construction:** a column packed with a **bed of solid particles** (porous or non-porous, rigid or compressible gels) with **immobilized** enzymes or cells on or inside them.
- **Working:** nutrient broth flows continuously through the bed (up or down); product leaves with the outflow. A **trickle bed** is a variant where liquid trickles down over the packing and air flows up.
- **Advantages:** high catalyst density, simple, continuous, high conversion, no need to separate catalyst from product.
- **Disadvantages:** poor mixing and O_2 supply, channelling, clogging, pH and heat gradients, compression of soft gels.
- **Uses:** immobilized glucose isomerase (HFCS), vinegar (trickling generator), effluent treatment.

(f) Membrane / hollow fibre bioreactor

- **Construction:** bundles of hollow semi-permeable fibres inside a shell. Cells are kept in the **shell (extracapillary) space**; medium flows through the fibres.
- **Working:** nutrients and oxygen diffuse out of the fibres to the cells; small products and wastes diffuse back into the flowing medium, while cells and large molecules are retained.
- **Advantages:** very high cell density, low shear, product separated continuously, cells protected.
- **Disadvantages:** membrane fouling, nutrient gradients, difficult scale-up, costly.
- **Uses:** animal cells, hybridomas (monoclonal antibodies), enzyme reactors, artificial organs.

(g) Photobioreactor

- **Definition:** a bioreactor with a **light source** for growing **photosynthetic** organisms (microalgae, cyanobacteria).
- **Open systems: raceway ponds** (shallow oval channels with a paddle wheel), cheap but easily contaminated, with poor control.
- **Closed systems: tubular** (transparent glass or plastic tubes), **flat-plate (panel)** and **column** reactors; good control of light, temperature, CO_2, and low contamination, but costly and can overheat.
- **Important factors:** light intensity and path (thin layers), CO_2 supply, O_2 removal (excess O_2 inhibits growth), temperature, mixing.
- **Uses:** *Spirulina*, *Chlorella* (SCP, health food), β-carotene (*Dunaliella*), astaxanthin (*Haematococcus*), algal biofuel.

3.5 Comparison of bioreactors

Bioreactor	Mixing by	Shear	Best for
Stirred tank (CSTR)	Impeller	High	Most bacterial and fungal fermentations
Bubble column	Rising air	Low	Yeast, simple aerobic processes
Airlift	Air + density difference (riser/downcomer)	Low	Shear-sensitive cells, SCP
Fluidized bed	Upward liquid flow	Medium	Immobilized cells, wastewater
Packed bed	Plug flow through bed	Low	Immobilized enzymes
Hollow fibre	Diffusion across membrane	Very low	Animal cells, antibodies
Photobioreactor	Air/pump + light	Low	Algae, cyanobacteria

Exam point: Always draw the stirred tank fermenter with at least 10 labels. For each other type write construction, working, advantages, disadvantages and uses. Airlift: remember riser, downcomer, draft tube, internal vs external loop.

4. TYPES OF FERMENTATION METHODS

Fermentations are classified by **mode of operation** (batch, fed-batch, continuous), by the **physical state of the medium** (submerged, surface, solid state, semi-solid), and by **oxygen requirement** (aerobic, anaerobic).

4.1 Batch fermentation (closed system)

- All nutrients are added at the start; the medium is sterilized and inoculated, and the fermentation runs to the end **without adding or removing** anything except air, acid/base for pH control and antifoam. At the end the whole broth is harvested.
- Cells pass through the **lag, log, stationary and death phases** (Fig. 1). Nutrients fall and products and waste build up, so conditions keep changing.
- **Kinetics:** in the log phase $dX/dt = \mu X$, where μ is the specific growth rate. Doubling time $t_d = 0.693/\mu$. The **Monod equation** relates μ to substrate: $\mu = \mu_{max} \cdot S / (K_s + S)$.
- **Advantages:** simple, less risk of contamination and strain mutation, flexible (one plant for many products), suits **secondary metabolites** made in stationary phase.
- **Disadvantages:** "down-time" between batches for cleaning, sterilizing and filling; low overall productivity; batch-to-batch variation.
- **Examples:** beer, wine, many antibiotics, enzymes, vaccines.

4.2 Fed-batch fermentation

- A batch culture that is **fed** with fresh nutrient (or precursor) continuously or at intervals during the run, **without removing broth**; the volume increases, and everything is harvested at the end.
- **Why used:** keeps substrate at a low level and so avoids **substrate inhibition** (e.g. high glucose), **catabolite repression** (glucose repressing secondary metabolism) and the **Crabtree effect** in yeast (ethanol formed at high sugar); allows toxic precursors (phenylacetic acid) to be added slowly; extends the production phase; gives high cell density.
- **Types:** variable-volume (volume increases) and fixed-volume (substrate fed as concentrated solution); feeding can be constant, exponential, or controlled by DO or pH (feedback).
- **Examples: penicillin**, baker's yeast, many recombinant proteins, amino acids.

4.3 Continuous fermentation (open system)

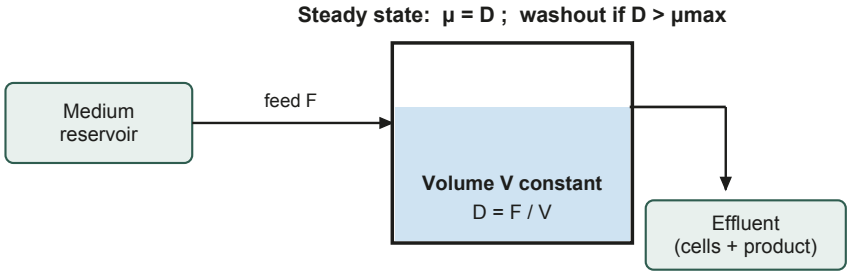


Fig. 4. Chemostat: fresh medium enters and broth leaves at the same rate

- Fresh medium is added **continuously** and culture broth is removed **at the same rate**, so the volume stays constant. After some time the culture reaches a **steady state** where cell, substrate and product concentrations stay constant and cells stay in the log phase.
- **Dilution rate** $D = F/V$ (F = flow rate, V = volume); units h^{-1}.
- Cell balance: $dX/dt = \mu X - DX$. At steady state $dX/dt = 0$, so **$\mu = D$**. The growth rate can therefore be set by the pump.
- If D is raised above μ_{max} (critical dilution rate), cells are removed faster than they grow and are lost: **washout**.

Type	Control	Notes
Chemostat	Constant flow of medium with one **growth-limiting nutrient**; growth rate set by D	Most common; stable at D below μ_{max}; used to study kinetics and physiology
Turbidostat	Cell density (turbidity) kept constant by a photocell that controls the feed pump	Works near μ_{max}; needs complex control
Multi-stage	Two or more vessels in series	First for growth, next for product formation (secondary metabolites)
With cell recycle	Cells separated (settling, membrane) and returned	Higher cell density and productivity; used in waste treatment and ethanol

- **Advantages:** high productivity, less down-time, uniform product, easier automation, cells kept in the most active state.
- **Disadvantages:** high risk of **contamination** over long runs, genetic instability and **back-mutation** of improved strains, not suitable for most secondary metabolites, wall growth, and difficult with filamentous organisms.
- **Examples:** SCP (Pruteen), ethanol, baker's yeast (some plants), vinegar, activated sludge process.

Feature	Batch	Fed-batch	Continuous
System	Closed	Semi-closed	Open
Volume	Constant	Increases	Constant
Nutrient addition	Only at start	During run	Continuous
Product removal	At end	At end	Continuous
Growth phase	All phases	Extended log / production	Steady log phase
Contamination risk	Low	Medium	High
Productivity	Low	High	Highest
Example	Beer, antibiotics	Penicillin, baker's yeast	SCP, ethanol

4.4 Submerged fermentation (SmF)

- Microorganisms grow **dispersed in a liquid medium** with aeration and agitation, in a stirred tank or airlift fermenter.
- **Advantages:** easy control of pH, temperature, O_2; uniform conditions; easy scale-up and automation; used for most industrial products.
- **Disadvantages:** high energy and water use, product is diluted, more effluent.
- **Examples:** antibiotics, most enzymes, citric acid (modern), vitamins, amino acids.

4.5 Surface (static) fermentation

- Organism grows as a **mat or pellicle on the surface** of a still liquid medium in shallow trays or pans; no agitation.
- **Examples:** early citric acid production by *A. niger* in aluminium trays, the Orleans method of vinegar, early penicillin in flasks.
- Now largely replaced by submerged fermentation because of low productivity and large space needs.

4.6 Solid state fermentation (SSF)

- Growth of microorganisms on **moist solid substrates** in the **absence or near absence of free-flowing water**. The solid gives both support and nutrients.
- **Substrates:** wheat bran, rice bran, rice straw, sugarcane bagasse, corn cobs, fruit pulp, oil cakes.
- **Organisms:** mainly **filamentous fungi** (*Aspergillus*, *Rhizopus*, *Trichoderma*, *Penicillium*), as they tolerate low water; some yeasts and bacteria.
- **Important factors:** moisture content (about 40–80%) and **water activity** (a_w), particle size, aeration, temperature (heat build-up is the main problem), pH, inoculum.

- **SSF bioreactors:** tray, packed bed (column), rotating drum, stirred and fluidized-bed reactors.
- **Examples: koji** (*Aspergillus oryzae* on rice/soybean for soy sauce and sake), tempeh (*Rhizopus*), mushroom cultivation, composting, enzymes (amylase, cellulase, pectinase, protease), gibberellic acid, fungal spores for biopesticides.

4.7 Semi-solid fermentation

- Intermediate between SmF and SSF: the substrate is a **thick slurry or paste** that has some free water but not enough to be a true liquid (e.g. bran with a higher water ratio).
- Combines some advantages of both: better mixing and heat removal than SSF, higher product concentration than SmF.
- Used for enzymes such as **lipase** and protease (syllabus practical: lipase by semi-solid fermentation).

Feature	Submerged (SmF)	Solid state (SSF)
Free water	Plenty	Little or none
Organisms	Bacteria, yeast, fungi	Mainly filamentous fungi
Aeration	Needs forced aeration and agitation	Air diffuses through spaces between particles
Control of pH, temperature	Easy and uniform	Difficult; gradients and heat build-up
Contamination	Bacterial contamination possible	Less (low water activity)
Product concentration	Low (dilute)	High (concentrated)
Energy, water, effluent	High	Low
Scale-up	Easy	Difficult
Examples	Penicillin, citric acid, enzymes	Koji, fungal enzymes, compost, mushrooms

4.8 Other classifications

- **Aerobic** (need O_2): antibiotics, citric acid, SCP, enzymes, vinegar. **Anaerobic**: ethanol, lactic acid, acetone–butanol, biogas.
- **Pure culture** vs **mixed culture** (curd, kefir, biogas, composting).

Exam point: Draw the growth curve and chemostat. Write μ = D at steady state and explain washout. Give the batch vs fed-batch vs continuous table and the SmF vs SSF table.

5. ENZYME PRODUCTION

5.1 Why microbial enzymes

- Microbes grow fast, give **high yields** in days, and can be grown in any season.
- Many enzymes are **extracellular**, so extraction is easy and cheap.
- Large variety of enzymes, including ones stable at high temperature (thermophiles), extreme pH (alkaliphiles) or in detergents.
- Yield can be increased by strain improvement and genetic engineering; cheap media.

5.2 Types of microbial enzymes

- **Extracellular** (exoenzymes): secreted into the medium, e.g. amylase, protease, lipase, cellulase, pectinase. Easy to recover.
- **Intracellular** (endoenzymes): stay inside cells, e.g. glucose isomerase, glucose oxidase, asparaginase, penicillin acylase. Cells must be broken.
- **Constitutive**: always made. **Inducible**: made only when an **inducer** (usually the substrate) is present, e.g. amylase by starch, cellulase by cellulose.

5.3 Regulation that affects yield

- **Induction**: add the inducer to the medium.
- **Catabolite repression**: glucose represses many enzymes, so use slowly used carbon sources or fed-batch feeding.
- **Feedback (end-product) repression**: product accumulation lowers synthesis; use regulatory mutants.
- Improve yield by mutants that are **constitutive** and **catabolite-derepressed**, and by cloning extra gene copies.

5.4 Steps in industrial enzyme production

1. **Selection of organism**: high yield, GRAS, stable, short fermentation, no toxins or antibiotics; e.g. *Bacillus*, *Aspergillus*, *Trichoderma*, *Streptomyces*, yeasts.
2. **Medium**: cheap carbon (starch, molasses, bran), nitrogen (soybean meal, CSL, ammonium salts), minerals, **inducer**; pH and C:N ratio optimized.
3. **Sterilization** of medium and fermenter.
4. **Inoculum**: spores or cells → shake flasks → seed tank (about 5–10% v/v).
5. **Fermentation**: SmF (most bacterial enzymes, 2–7 days) or SSF (many fungal enzymes); controlled temperature, pH, aeration; harvest when activity is highest (usually late log or early stationary).
6. **Downstream processing** (Fig. 5): separation, cell disruption (for intracellular), concentration, precipitation, purification.
7. **Formulation and standardization**: add stabilizers (glycerol, sorbitol, NaCl, calcium), preservatives; make liquid, powder or dust-free granules (for detergents); adjust to a fixed activity.

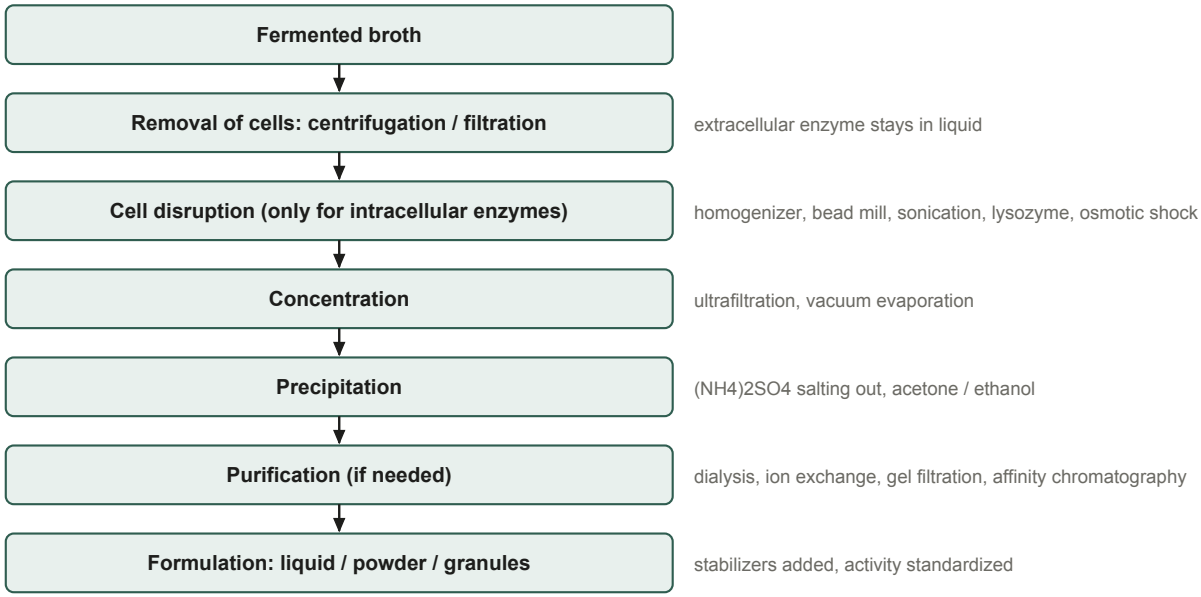


Fig. 5. Downstream processing of microbial enzymes

5.5 Examples of enzyme production

α-Amylase (bacterial)

- Organism: *Bacillus amyloliquefaciens*, *B. licheniformis* (thermostable, active at about 90°C), *B. subtilis*.
- Medium: starch (inducer), soybean meal or CSL, $CaCl_2$ (calcium stabilizes amylase); submerged, aerobic, about 30–40°C, pH about 7, 2–4 days.
- Recovery: remove cells, concentrate by ultrafiltration, precipitate or sell as stabilized liquid.
- Fungal amylase is made by *Aspergillus oryzae* in SSF (wheat bran) or SmF.
- Uses: starch liquefaction for glucose syrups, baking, brewing, textile desizing, detergents.

Protease (alkaline, detergent)

- Organism: *Bacillus licheniformis* (subtilisin Carlsberg), *B. subtilis*, alkalophilic *Bacillus*.
- Medium: starch, soybean meal, casein; submerged, aerobic, about 30–37°C, pH 7–9, 2–3 days.
- Recovery: filtration, concentration, precipitation, granulation (dust-free granules to avoid allergy).
- Uses: detergents (largest market), leather dehairing, meat tenderizing, silver recovery from X-ray film.

5.6 Enzyme activity and purity

- **Enzyme unit (U)**: the amount of enzyme that converts **1 µmol of substrate per minute** under defined conditions. SI unit: **katal** (1 mol/s).
- **Specific activity** = units per mg protein; rises during purification.
- **Purification fold** = specific activity after step ÷ specific activity of crude extract; **yield (%)** = total units recovered ÷ total units at start × 100.

5.7 Important microbial enzymes and applications

Enzyme	Source	Application
α-Amylase	*B. amyloliquefaciens*, *B. licheniformis*, *A. oryzae*	Starch liquefaction, baking, brewing, textile desizing, detergent
Glucoamylase	*Aspergillus niger*, *Rhizopus*	Glucose syrup, brewing, ethanol
Alkaline protease	*B. licheniformis*, *B. subtilis*	Detergents, leather, meat tenderizing
Lipase	*Candida rugosa*, *Rhizopus*, *Aspergillus*, *Pseudomonas*	Detergent, dairy flavour, fat modification, biodiesel
Cellulase	*Trichoderma reesei*, *Aspergillus*	Bioethanol, stone-washed jeans, juice extraction, feed

Enzyme	Source	Application
Pectinase	*Aspergillus niger*	Clarification of fruit juice and wine
Glucose isomerase	*Streptomyces*, *Actinoplanes*	High fructose corn syrup (HFCS)
Invertase	*Saccharomyces cerevisiae*	Invert sugar, soft-centre chocolates
Lactase (β-galactosidase)	*Kluyveromyces lactis*, *Aspergillus*	Lactose-free milk, whey use
Microbial rennet / chymosin	*Rhizomucor miehei*; recombinant in *K. lactis*, *A. niger*	Cheese making
Xylanase	*Trichoderma*, *Aspergillus*	Pulp bleaching, baking, feed
Penicillin acylase	*E. coli*, *Bacillus megaterium*	6-APA for semi-synthetic penicillins

Therapeutic enzymes

Enzyme	Source	Use
L-Asparaginase	*E. coli*, *Erwinia chrysanthemi*	Acute lymphoblastic leukaemia (removes asparagine needed by tumour cells)
Streptokinase	β-haemolytic *Streptococcus*	Dissolves blood clots (heart attack), activates plasminogen
Urokinase	Human kidney cells / recombinant *E. coli*	Thrombolytic (clot dissolving)
Serratiopeptidase	*Serratia marcescens*	Anti-inflammatory
Collagenase	*Clostridium histolyticum*	Wound debridement, burns
Lysozyme	Egg white / microbial	Antibacterial

Analytical enzymes

- **Glucose oxidase** (*A. niger*) + peroxidase: blood and urine glucose tests, glucose **biosensors** (glucometer strips).
- **Cholesterol oxidase** (*Streptomyces*, *Nocardia*): serum cholesterol test. **Urease**: blood urea. **Luciferase**: ATP detection (hygiene tests).
- **Horseradish peroxidase** and **alkaline phosphatase**: enzyme labels in **ELISA**.
- Molecular biology: **restriction endonucleases**, DNA ligase, **Taq polymerase** (*Thermus aquaticus*, PCR), reverse transcriptase.

Exam point: Answer order: why microbes → types (extra/intracellular, inducible) → steps with medium and conditions → downstream flow chart → one example (amylase or protease) → therapeutic / analytical / industrial applications table.

6. IMMOBILIZATION OF ENZYMES

6.1 Definition

Enzyme immobilization is the attachment or confinement of an enzyme (or whole cells) on or within an inert, insoluble **support (matrix)** so that it keeps its catalytic activity and can be **used repeatedly or continuously**. The first example was **invertase adsorbed on charcoal** (Nelson and Griffin, **1916**); the first industrial process was **aminoacylase** on DEAE-Sephadex for L-amino acids (**Chibata, Tanabe Seiyaku, Japan, 1969**).

6.2 Advantages

- **Reuse** of the enzyme many times, which lowers cost.
- Allows **continuous** processes in packed bed or fluidized bed reactors.
- Product is **not contaminated** by enzyme, so purification is easy.
- Often **greater stability** to temperature, pH, organic solvents and storage.
- Reaction can be **stopped quickly** by removing the enzyme; better process control.
- High enzyme concentration per reactor volume.

6.3 Disadvantages

- **Loss of activity** during immobilization (active site blocked or changed).
- **Diffusion (mass transfer) limitation**: substrate must reach the enzyme; big substrates move slowly.
- Extra cost of support and process.
- Leakage of enzyme from some supports; microbial contamination in long runs.
- Not useful when the substrate is insoluble or very large.

6.4 Properties of a good support

- Inert, non-toxic, insoluble, cheap and easily available.
- Mechanically strong, chemically and microbially stable.
- Large surface area, porous, allowing substrate and product to diffuse.
- Has groups for binding (-OH, -NH_2, -COOH); can be regenerated.

Type of support	Examples
Natural organic polymers	Alginate, agar, agarose, carrageenan, cellulose, chitosan, gelatin, collagen, starch
Synthetic organic polymers	Polyacrylamide, polystyrene, nylon, polyvinyl alcohol, ion-exchange resins
Inorganic materials	Activated charcoal, porous glass, silica, alumina, clay, bentonite

6.5 Methods of immobilization

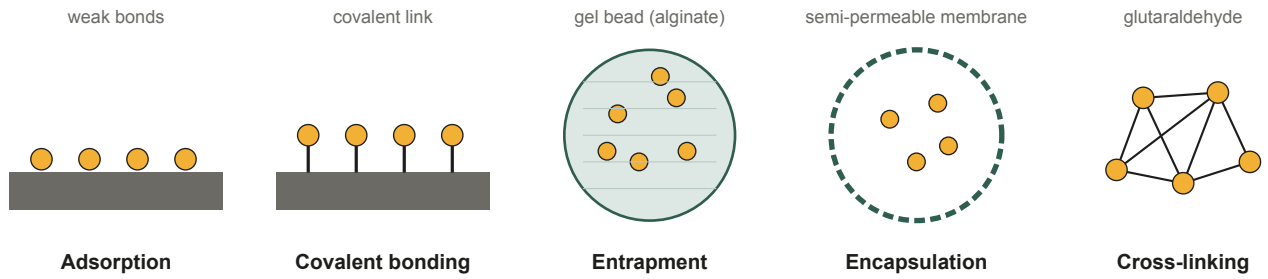


Fig. 6. Methods of enzyme immobilization (enzyme shown as yellow circles)

(1) Adsorption

- Enzyme is bound to the surface of the support by **weak forces**: van der Waals, ionic, hydrogen and hydrophobic bonds.

- **Procedure:** mix enzyme solution with the support under suitable pH and ionic strength, then wash off unbound enzyme.
- **Supports:** activated charcoal, alumina, silica, clay, glass, ion-exchange materials (DEAE-cellulose, CM-cellulose, DEAE-Sephadex).
- **Merits:** simple, cheap, mild, little change in enzyme, support can be reused. **Demerits:** enzyme **leaks (desorbs)** with change in pH, temperature or ionic strength.

(2) Covalent bonding

- Enzyme is joined by **covalent bonds** between its functional groups (-NH_2 of lysine, -COOH, -SH, -OH) and a **chemically activated** support. Groups needed for catalysis must not be used.
- **Activating agents:** cyanogen bromide (CNBr) for agarose/Sepharose and cellulose, **glutaraldehyde**, carbodiimide, silanes for glass.
- **Merits:** strong binding, **no leakage**, stable. **Demerits:** harsh chemistry can **inactivate** the enzyme; support cannot be reused; costly.

(3) Entrapment

- Enzyme or whole cells are **physically trapped** inside the lattice of a polymer gel or fibres; small substrates and products diffuse through the pores, the enzyme cannot escape.
- **Calcium alginate method (most common):** mix enzyme or cells with **2–3% sodium alginate** → drop through a syringe into **0.1–0.2 M $CaCl_2$** → Ca^{2+} cross-links alginate and forms **beads** → harden for 30 min, wash.
- Other gels: **polyacrylamide** (polymerized with bis-acrylamide), κ-carrageenan (with KCl), agar, gelatin, polyvinyl alcohol.
- **Merits:** mild, simple, suits **whole cells**, no chemical change of enzyme. **Demerits:** small enzymes can leak; diffusion limitation; alginate dissolves with phosphate or citrate (chelators).

(4) Encapsulation (microencapsulation)

- Enzyme is enclosed inside **semi-permeable membrane capsules** (microcapsules about 10–100 μm) or liposomes. Only small molecules pass through the membrane.
- **Membranes:** nylon, cellulose nitrate, polyethyleneimine; lipid vesicles (liposomes); hollow fibre membranes.
- **Merits:** large amount of enzyme held; several enzymes can be co-encapsulated. **Demerits:** only for small substrates; capsules may break.

(5) Cross-linking (carrier-free)

- Enzyme molecules are joined **to each other** by covalent bonds using **bifunctional reagents**, forming insoluble aggregates without any support.
- **Reagent: glutaraldehyde** (most common), hexamethylene diisocyanate; modern forms: CLEC (cross-linked enzyme crystals), **CLEA** (cross-linked enzyme aggregates).
- **Merits:** simple, no support cost, high enzyme density. **Demerits:** loss of activity, gel-like aggregates weak and hard to handle; often combined with adsorption.

Feature	Adsorption	Covalent	Entrapment	Encapsulation	Cross-linking
Binding force	Weak physical	Covalent	Physical trapping	Membrane barrier	Covalent (enzyme-enzyme)
Preparation	Simple	Difficult	Moderate	Moderate	Simple
Activity loss	Low	High	Low	Low	Moderate–high
Enzyme leakage	Yes	No	Possible	No (small enzymes may)	No
Cost	Low	High	Low	Moderate	Low

6.6 Changes in properties after immobilization

- **Stability** usually increases (thermal, pH, storage).

- **Optimum pH** may shift because of the charge of the support (microenvironment effect).
- **Apparent K_m** often increases because of diffusion limitation; V_{max} may fall.
- Activity towards large substrates falls (steric hindrance).

6.7 Immobilized whole cells

- Cells are immobilized (usually in alginate or carrageenan) when the enzyme is intracellular, unstable when purified, or needs cofactors.
- Advantages: no enzyme extraction, multi-step reactions possible, cofactors regenerated. Examples: immobilized *E. coli* (aspartase) for L-aspartic acid (Tanabe, 1973); *Brevibacterium* (fumarase) for L-malic acid; immobilized yeast for ethanol and beer; *Rhodococcus* (nitrile hydratase) for acrylamide.

6.8 Applications

Immobilized enzyme / cell	Product / use
Glucose isomerase (*Streptomyces*)	Glucose → fructose: **high fructose corn syrup**; the largest immobilized enzyme process
Aminoacylase (*Aspergillus oryzae*)	Resolution of DL-amino acids to L-amino acids (L-methionine, L-phenylalanine); first industrial process 1969
Penicillin G acylase (*E. coli*)	Penicillin G → **6-APA** for ampicillin, amoxicillin
Lactase (β-galactosidase)	Lactose-free milk; whey hydrolysis
Invertase	Invert sugar syrup
Nitrile hydratase (*Rhodococcus* cells)	Acrylonitrile → **acrylamide**
Aspartase (*E. coli* cells)	L-aspartic acid (for aspartame sweetener)
Glucose oxidase on electrode	**Biosensors** for blood glucose
Urease in microcapsules	Artificial kidney (removal of urea)

Exam point: Draw Fig. 6 and describe each method with principle, support/agent, merits and demerits. Mention the Ca-alginate bead procedure, glutaraldehyde, and glucose isomerase → HFCS as the main application.

7. SCREENING AND STRAIN IMPROVEMENT

7.1 Screening

Screening is the use of highly selective procedures to **detect and isolate** microorganisms of industrial interest from a large microbial population. It separates the few useful isolates from thousands of useless ones in the least time and cost. Sources are soil (richest source, especially for actinomycetes and fungi), water, sewage, compost, decaying food, extreme environments, and culture collections (ATCC, MTCC Chandigarh, NCIM Pune, NRRL).

7.2 Primary screening (qualitative)

Detects **which** organisms make the desired product. It is quick, simple and done on a large number of isolates, usually on agar plates.

(a) Crowded plate technique (for antibiotic producers)

- Soil is diluted and plated so that plates are **crowded** with 300–400 colonies.
- Colonies producing antibiotics show a **clear zone of inhibition** around them where other colonies do not grow.
- These colonies are picked and purified. The technique is simple but only detects producers active against neighbouring colonies.

(b) Test organism / cross-streak and agar plug methods

- **Crowded plate with test organism**: after growth the plate is overlaid with soft agar containing a test bacterium (e.g. *Staphylococcus aureus*); zones show antibiotic activity against it.
- **Cross-streak**: isolate streaked in one line; test organisms streaked at right angles after growth; inhibition near the isolate shows its **spectrum**.
- **Giant colony technique**: a single large colony is grown and tested against many organisms.

(c) Auxanography (for growth-factor producers)

- Used to find organisms that make **vitamins or amino acids**.
- Plate seeded with a test organism that **needs** that growth factor; isolates are spotted on it. A **zone of growth** of the test organism around an isolate shows it produces the growth factor.

(d) Indicator dye / plate assay methods

Product	Plate method
Organic acids	pH indicator dye (bromocresol green, bromothymol blue) or $CaCO_3$ in medium: yellow zone or clear zone of dissolved $CaCO_3$
Amylase	Starch agar flooded with iodine: clear zone on blue background
Protease	Skim milk or casein agar: clear zone
Lipase	Tributyrin agar: clear zone; or Rhodamine B–olive oil plate (orange fluorescence)
Cellulase	Carboxymethyl cellulose (CMC) agar stained with Congo red: clear zone
Phosphate solubilizers	Pikovskaya agar: clear halo

(e) Enrichment culture technique

- A medium is made that favours only the desired type, e.g. only the target compound as sole carbon source (for degraders), or selective temperature/pH (for thermophiles, alkaliphiles). Repeated subculturing enriches the wanted organism.

7.3 Secondary screening (quantitative)

Done on the few organisms selected in primary screening, usually in **shake flasks** or small fermenters. It tells **how much** product is made and whether the organism is worth developing.

- Measures **yield**, productivity and fermentation time; compares isolates.
- **Identifies** the organism (to check pathogenicity and whether the product is new or already known).

- Finds whether the product is new, and its chemical and biological properties (spectrum, toxicity).
- Determines optimum **pH, temperature, aeration, medium** components and the need for special conditions.
- Shows whether the product is intracellular or extracellular and whether the organism is genetically stable.
- Checks if the product is destroyed by the organism itself, and economics of the process.

7.4 Strain improvement

Wild isolates usually give very low yields. **Strain improvement** is the deliberate genetic change of a microorganism to make it better for industry.

Aims

- Increase **yield** and productivity; shorten fermentation time.
- Use of **cheaper** substrates; reduce unwanted by-products and foaming.
- Improve genetic stability, tolerance to product, and ease of downstream processing (e.g. extracellular product, pellet morphology).

(A) Mutation and selection (classical method)

Cells or spores are exposed to a **mutagen** to give about 90–99% kill; survivors are screened for improved mutants. Repeated rounds give stepwise improvement.

Mutagen	Type	Action
UV light (260 nm)	Physical	Forms **thymine dimers**; errors during repair (SOS) cause mutations
X-rays, γ-rays	Physical (ionizing)	Break DNA strands, cause deletions and rearrangements
NTG (N-methyl-N′-nitro-N-nitrosoguanidine)	Chemical, alkylating	Methylates guanine → GC to AT transitions; most potent, acts at replication fork
EMS (ethyl methane sulphonate)	Chemical, alkylating	Ethylates guanine → GC to AT transitions
Nitrous acid	Chemical, deaminating	Deaminates adenine and cytosine → transitions
Base analogues (5-bromouracil, 2-aminopurine)	Chemical	Incorporated in DNA and mispair
Acridine dyes (acridine orange, proflavine)	Chemical, intercalating	Insert between bases → frameshift mutations

Selection of mutants

- **Random screening**: test thousands of survivors in flasks for higher yield (used for penicillin).
- **Auxotrophic mutants**: need a growth factor; found by **replica plating** and enriched by penicillin enrichment. Block a pathway so an intermediate accumulates (e.g. homoserine auxotroph of *C. glutamicum* overproduces **lysine**).
- **Regulatory (analogue-resistant) mutants**: grown on a toxic **analogue** of the end product; resistant mutants often have enzymes insensitive to **feedback inhibition** and overproduce (e.g. AEC-resistant lysine producers).
- **Catabolite-derepressed** and **constitutive** mutants for enzymes; **revertants** of non-producing mutants that sometimes produce more.

(B) Recombination

- **Protoplast fusion**: cell walls removed by lysozyme (bacteria) or lytic enzymes (fungi) in an osmotic stabilizer (sucrose, mannitol, KCl) → protoplasts of two strains mixed with **polyethylene glycol (PEG)** → fusion → regeneration of cell wall on osmotically stabilized medium → selection of recombinants. Used for *Streptomyces*, *Penicillium*, yeasts.
- **Parasexual cycle** (Pontecorvo) in fungi without a sexual stage: heterokaryon → diploid → mitotic recombination → haploidization; used in *Aspergillus niger* and *P. chrysogenum*.
- **Transformation, transduction and conjugation** in bacteria; sexual hybridization in yeasts.

(C) Recombinant DNA technology and modern methods

- **Gene cloning**: extra copies of key genes, strong promoters, removal of repressors; expression of foreign genes in *E. coli*, *Bacillus*, yeast (insulin, chymosin, interferon).
- **Site-directed mutagenesis / protein engineering**: change a specific amino acid, e.g. oxidation-resistant subtilisin for bleach detergents.
- **Metabolic engineering**: block competing pathways and remove feedback control, e.g. riboflavin-overproducing *B. subtilis*.
- **Directed evolution**, **genome shuffling**, and genome editing (CRISPR).

7.5 Classic example: penicillin strain improvement

Strain	How obtained	Approx. yield
P. notatum (Fleming)	Original isolate, surface culture	About 2–20 units/mL
P. chrysogenum NRRL 1951	Natural isolate from a mouldy cantaloupe (Peoria)	About 100 units/mL (submerged)
NRRL 1951 B25	Single-spore selection	About 250 units/mL
X-1612	**X-ray** mutant	About 500 units/mL
Wis Q-176	**UV** mutant (University of Wisconsin); non-pigmented later mutants	About 900 units/mL
Modern industrial strains	Repeated mutation (NTG, UV) and selection	Over 50,000 units/mL (tens of g/L)

7.6 Preservation of industrial strains

- **Subculturing** on slants and storage at 4°C (short term); **overlaying with sterile mineral oil**.
- Storage as dried spores in **soil or silica gel** (fungi, actinomycetes).
- **Lyophilization (freeze-drying)**: cells frozen and dried under vacuum, stored for years.
- **Cryopreservation** in **liquid nitrogen (−196°C)** or at −80°C with glycerol or DMSO as cryoprotectant (best for long-term stability).

Exam point: Screening: define, then primary (crowded plate, auxanography, indicator plates, enrichment) and secondary (list 6 points). Strain improvement: aims → mutagens table → mutant selection → protoplast fusion → rDNA → penicillin example.

8. ANTIBIOTIC PRODUCTION

8.1 Introduction

Antibiotics are low-molecular-weight **secondary metabolites** produced by microorganisms that, at low concentrations, inhibit the growth of or kill other microorganisms (Waksman). About 70% of known antibiotics come from **actinomycetes** (mainly *Streptomyces*), about 20% from fungi and 10% from other bacteria (*Bacillus*).

Basis	Classes and examples
Producer	Fungi: penicillin, cephalosporin, griseofulvin; Actinomycetes: streptomycin, tetracycline, erythromycin, chloramphenicol; Bacteria: bacitracin, polymyxin
Spectrum	Narrow (penicillin G: Gram-positive), broad (tetracycline, chloramphenicol), antifungal (griseofulvin, nystatin)
Mode of action	Cell wall synthesis (penicillin), protein synthesis (streptomycin, tetracycline), nucleic acid synthesis (rifampicin), cell membrane (polymyxin), mitosis in fungi (griseofulvin)
Chemical structure	β-lactams (penicillins, cephalosporins), aminoglycosides (streptomycin), tetracyclines, macrolides (erythromycin), polyenes (nystatin)

8.2 General steps in antibiotic production

1. Strain selection and preservation (high-yield mutant).
2. Inoculum development: spores → shake flask → seed tanks.
3. Production in a sterile, aerated, agitated fermenter, usually **fed-batch**, with trophophase (growth) followed by idiophase (production).
4. Recovery: separation of mycelium, extraction (solvent or ion exchange), purification, crystallization.
5. Quality control: potency (bioassay), sterility, pyrogen and toxicity tests; formulation.

8.3 Penicillin

Introduction

- β-lactam antibiotic discovered by **Fleming (1928)**. Basic nucleus is **6-aminopenicillanic acid (6-APA)**: a thiazolidine ring fused with a **β-lactam ring**, with a variable side chain (R-CO-).
- Natural penicillins depend on the precursor in the medium: **penicillin G** (benzylpenicillin, side chain from **phenylacetic acid**) and **penicillin V** (phenoxymethylpenicillin, from **phenoxyacetic acid**, acid-stable, oral).
- Action: inhibits **peptidoglycan cross-linking** (binds transpeptidases / penicillin-binding proteins) in growing bacteria; mainly active on Gram-positive bacteria.

Biosynthesis (short)

L-α-aminoadipic acid + L-cysteine + L-valine → (ACV synthetase) → **δ-(L-α-aminoadipyl)-L-cysteinyl-D-valine (ACV tripeptide)** → (isopenicillin N synthase) → **isopenicillin N** → (acyltransferase exchanges side chain with **phenylacetyl-CoA**) → **penicillin G**.

Production process

Stage	Details
Organism	High-yielding mutants of *Penicillium chrysogenum* (descendants of NRRL 1951 / Wis Q-176)
Inoculum	Spores from lyophilized stock → spore suspension on rice or bran → shake flasks → 1–2 seed fermenters; about 5–10% inoculum, **pellet** growth preferred
Production medium	**Corn steep liquor** (N, growth factors, some phenylacetic acid), **lactose** (slowly hydrolysed, so avoids glucose catabolite repression) or glucose / sucrose fed slowly, ammonium salts, **$CaCO_3$** (buffer), KH_2PO_4, $MgSO_4$, **precursor phenylacetic acid** (fed gradually, as it is toxic above about 0.1%), vegetable oil / antifoam
Fermenter	Stirred tank, 40,000–200,000 L, stainless steel; **fed-batch**; sterile air
Conditions	Temperature **25–27°C**; pH **6.5** (kept 6.4–6.8); high aeration (0.5–1.0 vvm) and agitation; 5–7 days (up to 8)

Stage	Details
Phase 1: trophophase	First 1–2 days: rapid mycelial growth, lactose/glucose used, pH first falls then rises as NH_3 is released; little penicillin
Phase 2: idiophase	Growth slows; **penicillin formed**; continuous feeding of sugar, ammonia and **precursor**; pH kept near 6.5 with acid/alkali
Phase 3	Nutrients exhausted, autolysis of mycelium, pH rises, penicillin production stops; harvest before this

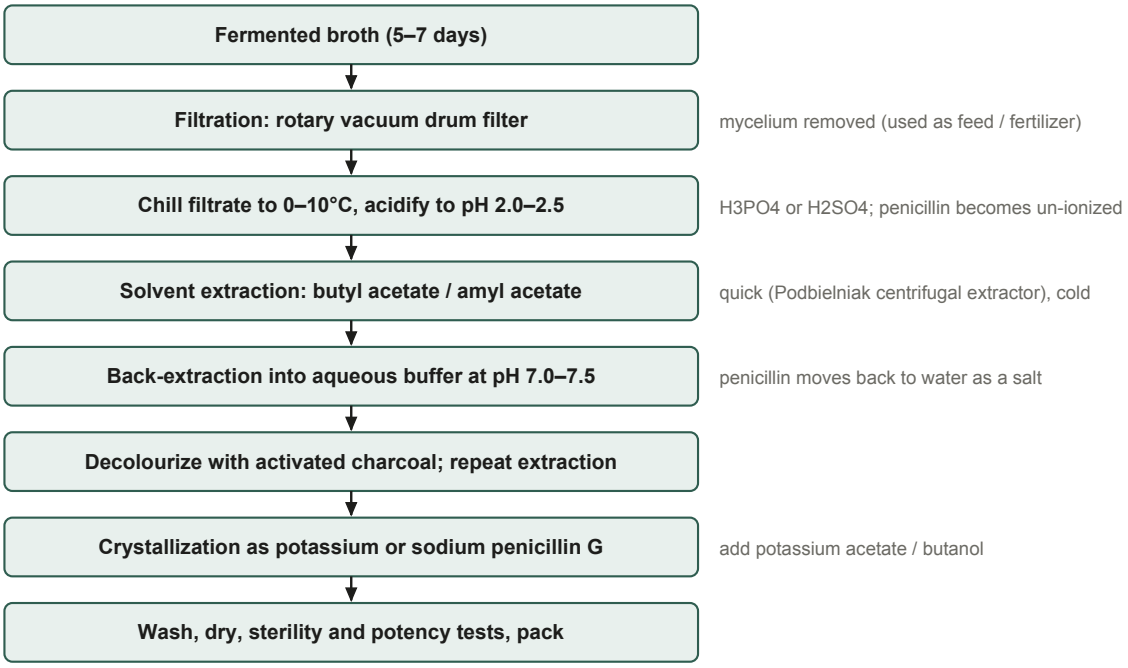


Fig. 7. Recovery (downstream processing) of penicillin

Semi-synthetic penicillins

- Penicillin G is split by **penicillin acylase** (immobilized *E. coli* enzyme) to give **6-APA** and phenylacetic acid.
- New side chains are joined chemically to 6-APA to make **ampicillin, amoxicillin, methicillin, cloxacillin**: broader spectrum, acid stability, or resistance to β-lactamase.

8.4 Streptomycin

- **Discovery:** Waksman, Schatz and Bugie, **1944**, from *Streptomyces griseus*.
- **Structure: aminoglycoside** made of three units: **streptidine** (from glucose via myo-inositol), **streptose** and **N-methyl-L-glucosamine**.
- **Action:** binds the **30S ribosomal subunit**, causes misreading of mRNA; active against Gram-negative bacteria and *Mycobacterium tuberculosis*.
- **Medium:** glucose, **soybean meal** (N source), NaCl, (CSL, distillers' solubles); a strain of *S. griseus* as spores → vegetative inoculum.
- **Conditions:** aerobic submerged, stirred tank, **about 28°C**, pH 7.6–8.0, high aeration, **about 7–10 days**.

Phase	What happens
Phase 1 (about 1 day)	Rapid growth of mycelium; proteolysis of soybean meal releases **NH_3**, so pH rises to about 8.0; glucose used slowly; little streptomycin
Phase 2 (days 2–6)	Little growth; **streptomycin accumulates**; glucose and NH_3 used; pH stays 7.6–8.0
Phase 3	Sugar exhausted; mycelium lyses; production stops; pH rises. Harvest at end of phase 2

- **Recovery:** acidify and filter off mycelium → adsorb streptomycin on **activated charcoal** or a **cation-exchange resin** → elute with dilute acid → concentrate → precipitate / crystallize as **streptomycin sulphate** (or calcium chloride complex) → dry.
- **By-product:** vitamin B_{12} can be recovered from the mycelium.

- **Uses:** tuberculosis (with other drugs), plague, tularaemia; in agriculture against plant bacterial diseases.

8.5 Tetracyclines

- **Chlortetracycline** (aureomycin, 1948, Duggar) from *Streptomyces aureofaciens*; **oxytetracycline** (terramycin) from *S. rimosus*; **tetracycline** from mutants of *S. aureofaciens* (or by removing Cl from chlortetracycline).
- **Structure:** four fused rings (naphthacene); a **polyketide** made from acetate/malonate units.
- **Action:** bind the **30S subunit** and block attachment of aminoacyl-tRNA; **broad spectrum** (Gram-positive, Gram-negative, rickettsiae, chlamydiae, mycoplasma).
- **Medium:** starch or sucrose, CSL, soybean meal, $CaCO_3$, $(NH_4)_2SO_4$; **phosphate kept low** (it represses production).
- **Key control:** to get **tetracycline** instead of chlortetracycline, **chloride is kept low** in the medium or a **chlorination inhibitor** (bromide, mercaptobenzothiazole) is added, or a chlorination-deficient mutant is used.
- **Conditions:** aerobic submerged, about 28°C, pH about 6.0–6.8, 4–6 days, fed-batch.
- **Recovery:** acidify broth (tetracycline binds Ca/Mg, so oxalic acid is added to release it), filter, extract with butanol or precipitate as a salt, purify and crystallize (as hydrochloride).

8.6 Griseofulvin

- **Antifungal** antibiotic from *Penicillium griseofulvum* (also *P. patulum*, *P. urticae*); discovered 1939 (Oxford, Raistrick).
- **Structure:** chlorine-containing **polyketide** (spiro benzofuran); a **chlorine atom** is part of the molecule.
- **Action:** binds **tubulin** and disrupts the mitotic spindle, stopping fungal cell division; deposited in keratin of skin, hair and nails.
- **Production:** aerobic submerged fed-batch fermentation, about 25°C, pH about 6–7, 7–10 days; medium with CSL, lactose or glucose, and **chloride (KCl)** added because it is incorporated into griseofulvin.
- **Recovery:** product is mostly in the mycelium: filter, extract mycelium with organic solvent (butyl acetate / acetone), concentrate and crystallize.
- **Uses:** oral treatment of **ringworm** (dermatophytes: *Trichophyton*, *Microsporum*, *Epidermophyton*) of skin, hair and nails; not active against bacteria or yeasts.

Antibiotic	Producer	Class	Target	Key production point
Penicillin G	*P. chrysogenum*	β-lactam	Cell wall	Phenylacetic acid precursor, lactose, fed-batch, pH 2–2.5 solvent extraction
Streptomycin	*S. griseus*	Aminoglycoside	30S ribosome	Soybean meal, 3 phases, cation exchange recovery
Tetracycline	*S. aureofaciens*, *S. rimosus*	Tetracycline (polyketide)	30S ribosome	Low chloride and low phosphate
Griseofulvin	*P. griseofulvum*	Polyketide (Cl)	Fungal tubulin	Chloride added; extracted from mycelium

Exam point: Penicillin is the most asked long question: organism → structure → inoculum → medium with precursor → fed-batch conditions → phases → recovery flow chart (Fig. 7) → semi-synthetic penicillins. Short notes on the other three from the summary table.

9. VITAMIN PRODUCTION

Vitamins are organic compounds needed in small amounts that the body cannot make. Some are made chemically; **riboflavin** and **vitamin B_{12}** are made mainly by **microbial fermentation** (B_{12} is made **only** by microbes).

9.1 Riboflavin (vitamin B_2)

- **Structure and role:** yellow, water-soluble vitamin (isoalloxazine ring + ribitol); forms the coenzymes **FMN** and **FAD** used in redox reactions. Deficiency: ariboflavinosis (cheilosis, glossitis).
- **Producers:** the ascomycete fungi ***Eremothecium ashbyii*** and ***Ashbya gossypii*** (natural overproducers, first used in the 1940s); the yeast *Candida famata* (flareri); and today mostly **genetically engineered *Bacillus subtilis*** and improved *A. gossypii*. Earlier, *Clostridium acetobutylicum* gave riboflavin as a by-product of acetone–butanol fermentation.
- **Biosynthesis:** from **GTP** and **ribulose-5-phosphate**, via 6,7-dimethyl-8-ribityllumazine; glycine and purine precursors increase yield.

Stage	Details (*Ashbya gossypii* process)
Inoculum	Stock → shake flasks → seed tank (about 5%)
Medium	Glucose or sucrose, or **plant oils** (soybean oil); **corn steep liquor**, peptone, collagen or gelatin, yeast extract; glycine (precursor); inositol; no iron limitation needed (unlike *Candida*, which needs **low iron**)
Conditions	Aerobic submerged, stirred tank, about **26–28°C**, pH 6–7, high aeration, about **5–7 days** (fed-batch)
Phases	Growth phase first; riboflavin made mostly after growth slows (**stationary phase**); much of it **crystallizes** in the cells and broth as yellow crystals
Recovery	Heat broth (about 1 h at 120°C) to release riboflavin from cells → centrifuge → collect crystals / precipitate → purify by recrystallization → dry. For feed grade, whole broth is dried
Yield	Modern processes give over 10–20 g/L
Uses	Animal feed supplement (most), human vitamin tablets, **yellow food colour** (E101)

9.2 Cyanocobalamin (vitamin B_{12})

- **Structure:** the largest and most complex vitamin: a **corrin ring** with a central **cobalt** atom, a lower ligand **5,6-dimethylbenzimidazole (DMB)** and an upper ligand (cyano-, methyl-, adenosyl- or hydroxo-). Commercial form is **cyanocobalamin**.
- **Role:** coenzyme in methionine synthase and methylmalonyl-CoA mutase; needed for red blood cell formation and nerves. Deficiency: **pernicious anaemia**.
- Made **only by bacteria and archaea**, not by plants or animals, so it is produced entirely by fermentation. First obtained as a by-product of streptomycin fermentation.
- **Producers:** ***Propionibacterium freudenreichii*** (and *P. shermanii*) and ***Pseudomonas denitrificans*** (main industrial organisms, highly improved strains); also *Bacillus megaterium*, *Streptomyces*.

Point	*Propionibacterium freudenreichii* process	*Pseudomonas denitrificans* process
Medium	Glucose, **corn steep liquor**, **cobalt** salt ($CoCl_2$), **DMB** added later	**Sugar beet molasses** (contains **betaine**, which strongly increases yield), CSL, cobalt, DMB
Conditions	**Two-stage**, about 30°C, pH about 7 (neutralized with NH_4OH)	Aerobic throughout, about 30°C, pH 6–7
Stage 1	**Anaerobic** for 2–4 days: growth and formation of the corrin precursor (cobinamide)	Single aerobic stage, fed-batch
Stage 2	**Aerobic** for 3–4 days: DMB is made (or added) and linked, forming complete B_{12}	—
Duration	About 6–7 days	About 5–7 days

- **Recovery:** B_{12} is **intracellular** → cells harvested by centrifugation → heated (80–120°C) at pH 6.5–8.5 to release the vitamin → **potassium cyanide** added to convert all cobalamins to stable **cyanocobalamin** → adsorption/solvent extraction and chromatography → crystallization. Feed grade: concentrated broth dried.
- **Uses:** treatment of **pernicious anaemia** and B_{12} deficiency, vitamin supplements (vegetarians), animal feed.

9.3 Other vitamins (extra marks)

- **Vitamin C (ascorbic acid)**: **Reichstein process**: D-glucose → (chemical hydrogenation) D-sorbitol → (***Gluconobacter oxydans*** oxidation) L-sorbose → chemical steps → ascorbic acid. The newer two-step fermentation uses *Gluconobacter* and *Ketogulonicigenium vulgare* to make 2-keto-L-gulonic acid.
- **β-Carotene (provitamin A)**: fungus *Blakeslea trispora* (mixed + and − mating strains), microalga *Dunaliella salina*.
- **Vitamin D**: ergosterol from yeast irradiated with UV gives vitamin D_2.

Exam point: Riboflavin: organisms (Ashbya, Eremothecium, B. subtilis) → medium → conditions → crystallization → recovery → uses. B_{12}: structure with cobalt and DMB → organisms → two-stage anaerobic–aerobic Propionibacterium process → betaine in Pseudomonas → cyanide in recovery → pernicious anaemia.

10. QUICK REVISION AND LIKELY QUESTIONS

Remember	Answer
Term "biotechnology"	Karl Ereky, 1919
Father of industrial microbiology	Louis Pasteur (fermentation by microbes, 1857)
Cell-free fermentation	Eduard Buchner, 1897 (zymase)
Acetone–butanol (WWI)	Chaim Weizmann, *Clostridium acetobutylicum*
Citric acid (industrial, 1923)	Pfizer, *Aspergillus niger* surface culture (Currie)
Penicillin	Fleming 1928; Florey and Chain; Peoria deep-tank, CSL, NRRL 1951
Streptomycin	Waksman, Schatz, Bugie 1944, *S. griseus*
First rDNA / first cloned human protein / first rDNA drug	Cohen–Boyer 1973 / somatostatin 1977 / Humulin 1982
Six components of fermentation	Media, sterilization, inoculum, production, downstream, effluent
Trophophase / idiophase	Growth phase (primary metabolites) / production phase (secondary metabolites)
Baffles	Usually 4, width 1/10 of diameter; prevent vortex
Airlift zones	Riser (gassed) and downcomer; draft tube; internal or external loop
Fluidized vs packed bed	Particles suspended by upward flow vs fixed bed
Chemostat	D = F/V; steady state $\mu = D$; washout if $D > \mu_{max}$
Monod equation	$\mu = \mu_{max} \cdot S / (K_s + S)$
Fed-batch avoids	Substrate inhibition, catabolite repression, Crabtree effect
SSF	Little free water, fungi, wheat bran, koji; a_w important
Enzyme unit	1 µmol substrate converted per minute; SI unit katal
First immobilized enzyme / first industrial	Invertase on charcoal 1916 / aminoacylase 1969 (Chibata)
Entrapment gel / cross-linker	Calcium alginate / glutaraldehyde
Largest immobilized enzyme use	Glucose isomerase → high fructose corn syrup
Crowded plate / auxanography	Antibiotic producers / growth-factor producers
Most potent chemical mutagen	NTG (GC→AT transitions)
Protoplast fusion agent	Polyethylene glycol (PEG)
Penicillin precursor / nucleus	Phenylacetic acid (Pen G) / 6-APA
Penicillin extraction	pH 2–2.5, butyl or amyl acetate, back-extract at pH 7.5
Streptomycin parts	Streptidine + streptose + N-methyl-L-glucosamine
Tetracycline control	Low chloride (or bromide) and low phosphate
Griseofulvin	*P. griseofulvum*; contains Cl; antifungal (tubulin); ringworm
Riboflavin producers	*Ashbya gossypii*, *Eremothecium ashbyii*, *B. subtilis*
B_{12}	*Propionibacterium* (anaerobic → aerobic), *Pseudomonas denitrificans* (betaine); cobalt + DMB; cyanide in recovery

Likely long questions

1. Define biotechnology. Trace the history of microbial biotechnology with important milestones.
2. What are the components of a fermentation process? Differentiate primary and secondary metabolites.
3. Describe the design and parts of a stirred tank fermenter with a labelled diagram.
4. Describe airlift, bubble column, fluidized bed, packed bed, hollow fibre and photobioreactors.
5. Differentiate batch, fed-batch and continuous fermentation. Explain the chemostat.

6 Compare submerged and solid state fermentation with examples.

7 Describe the industrial production of microbial enzymes and their therapeutic, analytical and industrial applications.

8 What is enzyme immobilization? Describe the methods with advantages and disadvantages and give applications.

9 Describe primary and secondary screening. Explain the methods of strain improvement.

10 Describe the industrial production and recovery of penicillin.

11 Write notes on production of streptomycin, tetracycline and griseofulvin.

12 Describe the microbial production of riboflavin and vitamin B_{12}.

Likely short notes (2–5 marks)

- Karl Ereky · Weizmann process · Peoria contribution · Buchner's experiment
- Baffles · sparger · Rushton turbine · $k_L a$ · washout · turbidostat
- Koji · water activity in SSF · enzyme unit · catabolite repression
- Calcium alginate beads · CLEA · glucose isomerase
- Crowded plate · auxanography · NTG · protoplast fusion · replica plating
- 6-APA · semi-synthetic penicillins · streptidine · chloride in tetracycline and griseofulvin
- Betaine and B_{12} · DMB · Ashbya gossypii

References: Stanbury, Whitaker and Hall, Principles of Fermentation Technology; Casida, Industrial Microbiology; Patel, Industrial Microbiology; Prescott and Dunn's Industrial Microbiology; Crueger and Crueger, Biotechnology: A Textbook of Industrial Microbiology; class notes "History of Biotechnology" and "Bioreactor Design, Unit 5" (Dr. Annam Renita A.).